

HIGH PERFORMANCE COMPUTING (HPC) + DEEP LEARNING VIVA QUESTIONS WITH ANSWERS

1) Parallel BFS and DFS using OpenMP

Basic Questions

Q1. What is BFS?

Answer:

Breadth First Search is a graph traversal algorithm that visits nodes level by level using a queue.

Q2. What is DFS?

Answer:

Depth First Search is a graph traversal algorithm that explores as deep as possible before backtracking. It generally uses recursion or stack.

Q3. Difference between BFS and DFS?

BFS	DFS
Uses Queue	Uses Stack/Recursion
Level-wise traversal	Depth-wise traversal
Finds shortest path	Does not guarantee shortest path
More memory	Less memory

Q4. Why did you use OpenMP?

Answer:

OpenMP is used for parallel programming in shared memory architecture. It helps execute loops concurrently using multiple threads.

Q5. What is #pragma omp parallel for?

Answer:

It divides loop iterations among multiple threads for parallel execution.

Q6. What data structure is used in BFS?

Answer:

Queue.

Q7. What data structure is used in DFS?

Answer:

Stack or recursion.

Q8. What is adjacency list?

Answer:

A graph representation where each vertex stores a list of connected vertices.

Q9. Why is visited[] array used?

Answer:

To avoid revisiting nodes and prevent infinite traversal.

Q10. Time complexity of BFS and DFS?

Answer:

$O(V + E)$

Where:

- V = Vertices
 - E = Edges
-

Code-Specific Questions**Q11. Why did you use vector in graph implementation?**

Answer:

Vector provides dynamic storage and efficient adjacency list representation.

Q12. Why recursion is used in DFS?

Answer:

Recursion naturally supports backtracking required in DFS traversal.

Q13. What happens if visited array is removed?

Answer:

The algorithm may revisit nodes repeatedly and enter infinite loops.

Q14. Is your BFS truly parallel?

Answer:

The neighbor processing loop is parallelized using OpenMP.

Q15. What is shared memory architecture?

Answer:

All processors share the same memory space.

Q16. What are race conditions?

Answer:

When multiple threads access and modify shared data simultaneously causing incorrect results.

Q17. Can BFS be implemented recursively?

Answer:

Generally BFS uses queue iteratively, recursion is uncommon.

Q18. Applications of BFS?

Answer:

- Shortest path
 - Social networks
 - Web crawling
 - Network broadcasting
-

Q19. Applications of DFS?

Answer:

- Cycle detection
 - Topological sorting
 - Maze solving
 - Backtracking
-

Q20. What is OpenMP?

Answer:

Open Multi-Processing is an API for parallel programming in C/C++ and Fortran.

2) Parallel Bubble Sort and Merge Sort using OpenMP

Bubble Sort Questions

Q1. What is Bubble Sort?

Answer:

Bubble Sort repeatedly swaps adjacent elements if they are in wrong order.

Q2. Time complexity of Bubble Sort?

Answer:

Worst case: $O(n^2)$

Q3. Why Bubble Sort is inefficient?

Answer:

Because it performs many unnecessary comparisons and swaps.

Q4. What is parallel Bubble Sort?

Answer:

Bubble sort where comparisons/swaps are divided among threads.

Q5. Explain odd-even transposition sort.

Answer:

It alternates between odd indexed and even indexed comparisons to enable parallelism.

Q6. Why #pragma omp barrier is used?

Answer:

To synchronize all threads before moving to next phase.

Q7. What is synchronization?

Answer:

Coordination between threads to maintain correct execution order.

Merge Sort Questions**Q8. Why Merge Sort is suitable for parallelism?**

Answer:

Because left and right halves can be sorted independently.

Q9. Time complexity of Merge Sort?

Answer:

$O(n \log n)$

Q10. What is divide and conquer?

Answer:

A technique that divides a problem into smaller subproblems and combines results.

Q11. What is #pragma omp sections?**Answer:**

It divides independent code blocks among threads.

Q12. Difference between sequential and parallel algorithms?

Sequential	Parallel
Single processor	Multiple processors
Slower	Faster
One task at a time	Multiple tasks simultaneously

Q13. Why Merge Sort is faster than Bubble Sort?**Answer:**

Merge Sort has lower complexity $O(n \log n)$.

Q14. What is speedup?**Answer:**

$\text{Speedup} = \text{Sequential Time} / \text{Parallel Time}$

Q15. What is efficiency?**Answer:**

$\text{Efficiency} = \text{Speedup} / \text{Number of processors}$

3) Parallel Reduction

Based on your reduction code

Q1. What is reduction in OpenMP?**Answer:**

Reduction combines results from multiple threads into a single variable.

Q2. Why reduction is used?**Answer:**

To safely perform operations like sum, min, max in parallel.

Q3. Syntax of reduction?

#pragma omp parallel for reduction(+:sum)

Q4. Which reduction operators are supported?

Answer:

+, *, min, max, &, | etc.

Q5. Why reduction avoids race conditions?

Answer:

Each thread uses private copy and combines results at the end.

Q6. Difference between shared and private variables?

Shared

Private

Accessible by all threads Separate copy per thread

Q7. What is race condition?

Answer:

Simultaneous modification of shared variable causing incorrect output.

Q8. Why average uses sum?

Answer:

Average = Sum / Number of elements

Q9. What is parallel overhead?

Answer:

Extra cost due to thread creation, synchronization, communication.

4) CUDA Programs**Vector Addition****Q1. What is CUDA?**

Answer:

CUDA is NVIDIA's parallel computing platform for GPU programming.

Q2. Why GPU is faster?

Answer:

GPU has thousands of cores for massive parallelism.

Q3. What is kernel function?

Answer:

Function executed on GPU.

Q4. What is thread in CUDA?

Answer:

Smallest execution unit in GPU.

Q5. What is block?

Answer:

Group of threads.

Q6. What is grid?

Answer:

Collection of thread blocks.

Q7. Why vector addition is parallelizable?

Answer:

Each element addition is independent.

Q8. What is __global__?

Answer:

CUDA keyword indicating GPU kernel function.

Matrix Multiplication

Q9. Why matrix multiplication is computationally expensive?

Answer:

It requires large number of multiply-add operations.

Q10. Complexity of matrix multiplication?

Answer:

$O(n^3)$

Q11. Applications of CUDA?

Answer:

- AI
 - Gaming
 - Scientific simulations
 - Image processing
-

5) Boston Housing Price Prediction using DNN

Basic Questions

Q1. What is Deep Learning?

Answer:

A subset of machine learning using multi-layer neural networks.

Q2. What is DNN?

Answer:

Deep Neural Network containing multiple hidden layers.

Q3. Why this is regression problem?

Answer:

Because output is continuous house price.

Q4. What dataset is used?

Answer:

Boston Housing Dataset.

Q5. What is feature scaling?

Answer:

Normalizing features to similar range.

Q6. Why ReLU activation is used?

Answer:

It reduces vanishing gradient problem and improves training speed.

Q7. What optimizer is used?

Answer:

Adam optimizer.

Q8. What is loss function?

Answer:

Function measuring prediction error.

Q9. Which loss function is used?

Answer:

Mean Squared Error (MSE)

Q10. What is epoch?

Answer:

One complete pass through training dataset.

Q11. What is batch size?

Answer:

Number of samples processed before updating weights.

Q12. Why output layer has one neuron?

Answer:

Because only one continuous value is predicted.

Q13. What is overfitting?

Answer:

Model performs well on training but poorly on testing data.

Q14. What is underfitting?

Answer:

Model fails to learn patterns properly.

Q15. Difference between AI, ML and DL?

AI	ML	DL
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Broad field	Subset of AI	Subset of ML
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6) IMDB Sentiment Classification using DNN

Q1. Why classification problem?

Answer:

Because output categories are positive or negative.

Q2. What is binary classification?

Answer:

Classification into two classes.

Q3. What dataset is used?

Answer:

IMDB movie reviews dataset.

Q4. Why text preprocessing is needed?

Answer:

To convert text into numerical format.

Q5. What is tokenization?

Answer:

Splitting text into words/tokens.

Q6. Which activation function is used in output layer?

Answer:

Sigmoid.

Q7. Why sigmoid?

Answer:

It outputs values between 0 and 1 for probability.

Q8. Which loss function is used?

Answer:

Binary Cross Entropy.

Q9. What is embedding?

Answer:

Dense vector representation of words.

Q10. Applications?

Answer:

- Sentiment analysis
 - Spam detection
 - Review analysis
-

7) CNN – Fashion MNIST

Q1. What is CNN?

Answer:

Convolutional Neural Network specialized for image processing.

Q2. Why CNN for images?

Answer:

CNN automatically extracts image features.

Q3. What is convolution?

Answer:

Applying filter/kernel to extract features.

Q4. What is filter/kernel?

Answer:

Small matrix used for feature extraction.

Q5. What is pooling?

Answer:

Reducing feature map size.

Q6. Types of pooling?

Answer:

- Max pooling
 - Average pooling
-

Q7. What is flatten layer?

Answer:

Converts 2D data into 1D vector.

Q8. What activation function commonly used?

Answer:

ReLU

Q9. What dataset used?

Answer:

Fashion MNIST.

Q10. Difference between CNN and ANN?

ANN

CNN

Fully connected

Uses convolution

Not ideal for images

Best for images

8) RNN – Google Stock Prediction

Q1. What is RNN?

Answer:

Recurrent Neural Network designed for sequential data.

Q2. Why RNN for stock prediction?

Answer:

Because stock prices are time-series sequential data.

Q3. What is time series data?

Answer:

Data collected over time intervals.

Q4. What is hidden state in RNN?

Answer:

Memory storing previous information.

Q5. Problem with basic RNN?

Answer:

Vanishing gradient problem.

Q6. What is LSTM?

Answer:

Long Short-Term Memory network solving vanishing gradient issue.

Q7. Applications of RNN?**Answer:**

- Stock prediction
 - Speech recognition
 - NLP
 - Translation
-

Q8. Difference between CNN and RNN?

CNN	RNN
Image data	Sequential data
Spatial features	Temporal features

VERY IMPORTANT COMMON EXTERNAL QUESTIONS**Q. Why parallel computing is needed?****Answer:**

To improve execution speed and solve large computational problems efficiently.

Q. Difference between process and thread?

Process	Thread
Independent execution	Lightweight execution unit
Separate memory	Shared memory

Q. What is multicore processor?**Answer:**

Processor containing multiple CPU cores.

Q. What is scalability?**Answer:**

Ability of system to improve performance with more processors.

Q. What is Amdahl's Law?

Answer:

It defines maximum speedup achievable using parallelism.

$$Speedup = \frac{1}{(1 - P) + \frac{P}{N}}$$

Where:

- P = Parallel portion
 - N = Number of processors
-

Q. Difference between shared memory and distributed memory?

Shared Memory	Distributed Memory
Common memory	Separate memory
Easier communication	Message passing required

Q. What is OpenMP?**Answer:**

API for shared-memory parallel programming.

Q. What is CUDA?**Answer:**

GPU programming platform by NVIDIA.

Q. Difference between CPU and GPU?

CPU	GPU
Few powerful cores	Thousands of small cores
Sequential tasks	Parallel tasks

Q. What is machine learning?**Answer:**

Technique where systems learn patterns from data without explicit programming.

Q. What is neural network?**Answer:**

Computational model inspired by biological neurons.

PRACTICAL 1

Parallel BFS and DFS using OpenMP

BASIC THEORY QUESTIONS

Q1. What is graph traversal?

Answer:

Graph traversal means visiting all nodes/vertices of a graph systematically.

Q2. What is BFS?

Answer:

Breadth First Search traverses the graph level by level using a queue.

Q3. What is DFS?

Answer:

Depth First Search traverses deeply along one branch before backtracking.

Q4. Difference between BFS and DFS?

BFS	DFS
Level-wise traversal	Depth-wise traversal
Uses Queue	Uses Stack/Recursion
Finds shortest path	Does not guarantee shortest path

Q5. Why use OpenMP?

Answer:

OpenMP enables parallel execution using multiple CPU threads.

Q6. What is #pragma omp parallel for?

Answer:

It divides loop iterations among multiple threads for parallel execution.

VERY IMPORTANT DRY RUN QUESTION

Q7. Explain BFS traversal with example.

Suppose graph:

$0 \rightarrow 1,2$

$1 \rightarrow 3,4$

$2 \rightarrow 5,6$

Start BFS from 0.

SOLUTION

Step-by-step

Initially

Queue:

[0]

Visited:

0

Remove 0

Output:

0

Insert neighbors:

1,2

Queue:

[1,2]

Remove 1

Output:

0 1

Insert:

3,4

Queue:

[2,3,4]

Remove 2

Output:

0 1 2

Insert:

5,6

Queue:

[3,4,5,6]

Final BFS

0 1 2 3 4 5 6

VERY IMPORTANT DRY RUN QUESTION

Q8. Explain DFS traversal with example.

Same graph:

0 → 1,2

1 → 3,4

2 → 5,6

SOLUTION

Start:

0

Go deep:

0 → 1 → 3

Backtrack:

4

Then:

2 → 5 → 6

Final DFS

0 1 3 4 2 5 6

CODE QUESTIONS

Q9. Why visited array is used?

Answer:

To avoid revisiting nodes and infinite loops.

Q10. Why queue used in BFS?

Answer:

Because BFS follows FIFO order.

Q11. Why recursion used in DFS?**Answer:**

Recursion naturally supports backtracking.

Q12. Time complexity of BFS and DFS?**Answer:**

$$O(V + E)$$

Where:

- V = vertices
 - E = edges
-

Q13. What is adjacency list?**Answer:**

A graph representation storing neighbors of each vertex.

Q14. What is parallelism in your code?**Answer:**

Neighbor traversal loop executes in parallel using OpenMP threads.

Q15. What is race condition?**Answer:**

Simultaneous modification of shared data by multiple threads causing incorrect output.

Q16. Why OpenMP preferred?**Answer:**

Simple syntax and efficient shared-memory parallelism.

Q17. Difference between process and thread?

Process	Thread
Heavyweight	Lightweight
Separate memory	Shared memory

PRACTICAL 2**Parallel Bubble Sort and Merge Sort**

BUBBLE SORT QUESTIONS

Q1. What is Bubble Sort?

Answer:

Bubble Sort repeatedly swaps adjacent elements if they are in wrong order.

Q2. Time complexity of Bubble Sort?

$$O(n^2)$$

Q3. Why Bubble Sort is inefficient?

Answer:

Large number of comparisons and swaps.

Q4. What is parallel Bubble Sort?

Answer:

Bubble sort where comparisons are distributed among multiple threads.

Q5. Explain odd-even transposition sort.

Answer:

- Odd indexed elements sorted first
 - Then even indexed elements
 - Enables parallel execution
-

DRY RUN

Array:

5 4 3 2 1

Pass 1

Compare:

5-4

4-3

3-2

2-1

After swaps:

4 3 2 1 5

Pass 2

3 2 1 4 5

Eventually:

1 2 3 4 5

MERGE SORT QUESTIONS

Q6. What is Merge Sort?

Answer:

Divide-and-conquer sorting algorithm.

Q7. Time complexity of Merge Sort?

$O(n \log n)$

Q8. Why Merge Sort better than Bubble Sort?

Answer:

Lower complexity and better performance.

Q9. Why Merge Sort suitable for parallelism?

Answer:

Left and right halves can be processed independently.

Q10. Explain divide and conquer.

Answer:

Problem divided into smaller subproblems, solved recursively and merged.

DRY RUN

Array:

8 4 2 6

Split:

8 4 | 2 6

Further:

8 | 4

2 | 6

Sort:

4 8

2 6

Merge:

2 4 6 8

Q11. What is #pragma omp sections?

Answer:

Executes independent sections in parallel.

Q12. Difference between sequential and parallel execution?

Sequential	Parallel
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One task at a time	Multiple tasks simultaneously
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Q13. What is speedup?

$$Speedup = \frac{Sequential\ Time}{Parallel\ Time}$$

Q14. What is efficiency?

$$Efficiency = \frac{Speedup}{Number\ of\ Processors}$$

PRACTICAL 3

Parallel Reduction

Q1. What is reduction?

Answer:

Combining partial results from multiple threads into one final result.

Q2. Why reduction required?

Answer:

To safely compute operations like sum/min/max in parallel.

Q3. Syntax of reduction?

#pragma omp parallel for reduction(+:sum)

Q4. What operations used?

Answer:

- Sum
 - Min
 - Max
 - Average
-

DRY RUN

Array:

1 2 3 4 5

Sum

$$1 + 2 + 3 + 4 + 5 = 15$$

Average

$$Average = \frac{15}{5} = 3$$

Min

1

Max

5

Q5. What is race condition?

Answer:

Incorrect results caused by simultaneous shared variable updates.

Q6. How reduction avoids race condition?

Answer:

Each thread maintains private copy then combines results.

Q7. Difference between shared and private variable?

Shared

Private

Accessible to all threads Separate per thread

PRACTICAL 4

CUDA — Vector Addition and Matrix Multiplication

CUDA QUESTIONS

Q1. What is CUDA?

Answer:

Parallel computing platform by NVIDIA for GPU programming.

Q2. Why GPU faster than CPU?

Answer:

GPU contains thousands of smaller cores enabling massive parallelism.

Q3. What is kernel function?

Answer:

Function executed on GPU.

Q4. What is __global__?

Answer:

CUDA keyword specifying GPU kernel.

Q5. What is thread?

Answer:

Smallest execution unit in CUDA.

Q6. What is block?

Answer:

Group of threads.

Q7. What is grid?

Answer:

Collection of thread blocks.

VECTOR ADDITION DRY RUN

Suppose:

$$A = [1, 2, 3]$$

$$B = [4, 5, 6]$$

Output

$$C = A + B = [5, 7, 9]$$

Q8. Why vector addition suitable for parallelism?

Answer:

Each addition operation is independent.

MATRIX MULTIPLICATION DRY RUN

Matrix A:

1 2

3 4

Matrix B:

5 6

7 8

Result

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \times \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Q9. Complexity of matrix multiplication?

$$O(n^3)$$

Q10. Applications of CUDA?

Answer:

- AI
- Gaming
- Scientific simulations
- Image processing
- Deep learning

PRACTICAL 5

Boston Housing Price Prediction using Deep Neural Network

AIM EXPLANATION

Q1. What is the aim of this practical?

Answer:

To predict house prices using a Deep Neural Network based regression model.

BASIC THEORY QUESTIONS

Q2. Why is this a regression problem?

Answer:

Because output is a continuous numerical value (house price).

Q3. What dataset is used?

Answer:

Boston Housing Dataset.

Q4. What does the dataset contain?

Answer:

Housing features like:

- Crime rate
- Number of rooms
- Tax rate
- Pollution level
- etc.

used to predict house prices.

Q5. How many features are there?

13 features

Q6. What is Deep Learning?

Answer:

A subset of machine learning using multiple neural network layers.

Q7. What is DNN?

Answer:

Deep Neural Network containing multiple hidden layers.

DATA PREPROCESSING QUESTIONS

Q8. Why feature scaling is required?

Answer:

Features may have different ranges. Scaling improves training stability and convergence speed.

Q9. Which scaling technique is used?

Answer:

StandardScaler / normalization.

Q10. Why train-test split is used?

Answer:

To evaluate model performance on unseen data.

MODEL QUESTIONS

Q11. What is Sequential model?

Answer:

A neural network where layers are arranged sequentially.

Q12. What is Dense layer?

Answer:

A fully connected neural network layer.

Q13. Why ReLU activation used?

$$ReLU(x) = \max(0, x)$$

Answer:

- Faster training
 - Solves vanishing gradient problem
 - Introduces non-linearity
-

Q14. Why output layer has 1 neuron?

Answer:

Because only one continuous output (house price) is predicted.

Q15. Which optimizer used?

Answer:

Adam optimizer.

Q16. What is optimizer?

Answer:

Algorithm that updates neural network weights during training.

Q17. Which loss function used?

Answer:

Mean Squared Error (MSE)

Q18. Formula for MSE

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Q19. What is epoch?

Answer:

One complete pass through training dataset.

Q20. What is batch size?

Answer:

Number of samples processed before updating weights.

OUTPUT INTERPRETATION QUESTIONS

Q21. What does MAE mean?

Answer:

Mean Absolute Error measures average prediction error.

Q22. Difference between MAE and MSE?

MAE	MSE
Absolute error	Squared error
Less sensitive to outliers	More sensitive to outliers

Q23. What is overfitting?

Answer:

High training accuracy but poor testing performance.

Q24. What is underfitting?

Answer:

Model fails to learn patterns properly.

VERY IMPORTANT

Q25. How does this practical fulfill the aim?

Answer:

“The model learns relationships between housing features and prices using a Deep Neural Network and predicts continuous house prices successfully.”

PRACTICAL 6

IMDB Binary Classification using DNN

VERY IMPORTANT THEORY QUESTIONS

Q1. Why is this binary classification?

Answer:

Because there are only two classes:

- Positive review
 - Negative review
-

Q2. What dataset is used?

Answer:

IMDB Movie Review Dataset.

Q3. What is sentiment analysis?

Answer:

Classifying text according to emotion/opinion.

Q4. Why preprocessing required?

Answer:

Neural networks require numerical input, not raw text.

Q5. What is vectorization?

Answer:

Converting text into numerical vectors.

IMPORTANT CONCEPT

Q6. What does num_words=10000 mean?

Answer:

Only top 10,000 most frequent words are used.

Q7. What activation used in output layer?

Answer:

Sigmoid activation.

Q8. Formula of sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Q9. Why sigmoid used?

Answer:

Because it outputs probabilities between 0 and 1 suitable for binary classification.

Q10. What do labels mean?

Label	Meaning
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1	Positive
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0	Negative
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Q11. What is embedding?

Answer:

Dense vector representation of words.

Q12. What is NLP?

Answer:

Natural Language Processing.

Q13. Which loss function is ideal?

Answer:

Binary Cross Entropy.

Q14. Formula for Binary Cross Entropy

$$L = -[y \log(p) + (1 - y) \log(1 - p)]$$

Q15. Why ReLU in hidden layers?

Answer:

Improves learning and reduces vanishing gradient problem.

Q16. Explain complete flow of practical.

Answer:

1. Load IMDB dataset
 2. Convert reviews into numerical vectors
 3. Build DNN model
 4. Train using labeled reviews
 5. Predict positive/negative sentiment
-

Q17. How does practical fulfill aim?

Answer:

“The DNN model successfully classifies movie reviews into positive and negative sentiments using the IMDB dataset.”

PRACTICAL 7

CNN using Fashion MNIST

BASIC QUESTIONS

Q1. What is CNN?

Answer:

Convolutional Neural Network specialized for image processing.

Q2. Why CNN used for images?

Answer:

CNN automatically extracts image features efficiently.

Q3. What is convolution?

Answer:

Applying filters to extract image features.

Q4. What is filter/kernel?

Answer:

Small matrix used for feature extraction.

VERY IMPORTANT

Q5. Explain convolution with example.

Suppose filter:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Applied on image region:

$$\begin{bmatrix} 5 & 2 \\ 1 & 3 \end{bmatrix}$$

Output:

$$(5 \times 1) + (2 \times 0) + (1 \times 0) + (3 \times 1) = 8$$

Q6. What is pooling?

Answer:

Reducing feature map size.

Q7. Types of pooling?

Type	Description
Max pooling	Takes maximum value
Average pooling	Takes average

Q8. What is flatten layer?

Answer:

Converts 2D feature maps into 1D vector.

Q9. What dataset used?

Answer:

Fashion MNIST dataset.

Q10. Why Fashion MNIST popular?

Answer:

Simple benchmark dataset for image classification.

Q11. Difference between CNN and ANN?

ANN

CNN

Fully connected

Uses convolution

Not ideal for images

Best for images

Q12. What activation used?

Answer:

ReLU.

Q13. Applications of CNN?

Answer:

- Face recognition
 - Medical imaging
 - Self-driving cars
 - Object detection
-

PRACTICAL 8

RNN using Google Stock Prediction

BASIC QUESTIONS

Q1. What is RNN?

Answer:

Recurrent Neural Network designed for sequential/time-series data.

Q2. Why RNN for stock prediction?

Answer:

Stock prices are sequential and time-dependent.

Q3. What is time-series data?

Answer:

Data collected over time intervals.

Q4. What is hidden state?**Answer:**

Memory storing previous information in RNN.

Q5. What is vanishing gradient problem?**Answer:**

Gradients become very small during backpropagation causing poor learning.

Q6. What is LSTM?**Answer:**

Long Short-Term Memory network that solves vanishing gradient problem.

Q7. Difference between ANN, CNN and RNN?

ANN	CNN	RNN
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General tasks	Images	Sequential data
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Q8. Applications of RNN?**Answer:**

- Stock prediction
 - Speech recognition
 - Language translation
 - NLP
-

Q9. What is sequence dependency?**Answer:**

Current output depends on previous inputs.

Q10. Why memory important in RNN?**Answer:**

Past information influences future predictions.

VERY IMPORTANT**Q11. How does practical fulfill aim?****Answer:**

“The RNN model learns patterns from historical stock prices and predicts future stock values using sequential dependencies.”

COMMON CROSS QUESTIONS (VERY IMPORTANT)

Q1. Difference between AI, ML and DL?

AI	ML	DL
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Broad field	Subset of AI	Subset of ML
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Q2. Difference between regression and classification?

Regression	Classification
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Continuous output	Categorical output
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Q3. What is activation function?

Answer:

Function introducing non-linearity into neural networks.

Q4. Why non-linearity required?

Answer:

Real-world data relationships are non-linear.

Q5. What is forward propagation?

Answer:

Passing inputs through neural network to generate output.

Q6. What is backpropagation?

Answer:

Updating weights using error gradients.

Q7. What is gradient descent?

Answer:

Optimization algorithm minimizing loss function.

Q8. Difference between CPU and GPU?

CPU

GPU

Few powerful cores Thousands of smaller cores

Q9. What is parallel computing?

Answer:

Executing multiple computations simultaneously.

Q10. What is scalability?

Answer:

Ability to improve performance with more processors.

Q11. What is Amdahl's Law?

$$Speedup = \frac{1}{(1 - P) + \frac{P}{N}}$$

Where:

- P = parallel portion
 - N = processors
-

Q12. Difference between OpenMP and CUDA?

OpenMP

CUDA

CPU parallelism GPU parallelism

Q13. What is overfitting solution?

Answer:

- More data
 - Dropout
 - Regularization
 - Early stopping
-

Q14. Why train-test split important?

Answer:

To evaluate generalization capability.

Q15. What is accuracy?

Answer:

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions}$$

EXTRA HIGH-LEVEL / TRICK QUESTIONS THEY MAY ASK

HPC SIDE

Q. Why parallel computing instead of increasing CPU frequency?

Answer:

Increasing frequency increases heat and power consumption, while parallel computing improves performance efficiently using multiple cores.

Q. What is multicore processor?

Answer:

A processor containing multiple CPU cores capable of parallel execution.

Q. What is shared memory architecture?

Answer:

All processors access the same common memory.

Q. Difference between distributed and shared memory?

Shared Memory	Distributed Memory
Common memory	Separate memory
Faster communication	Message passing required

Q. What is synchronization?

Answer:

Coordination between threads to maintain correct execution order.

Q. Why barriers are used in OpenMP?

Answer:

To ensure all threads complete before proceeding further.

Q. What is deadlock?

Answer:

Situation where processes wait indefinitely for resources.

Q. What is load balancing?

Answer:

Equal distribution of workload among processors.

Q. Why Merge Sort parallelizes better than Bubble Sort?

Answer:

Merge Sort divides independent subproblems naturally.

Q. What is thread scheduling?

Answer:

Allocation of threads to processors by operating system/runtime.

Q. What is CUDA memory hierarchy?

Answer:

Registers → Shared Memory → Global Memory → Local Memory.

Q. Difference between host and device in CUDA?

Host	Device
------	--------

CPU	GPU
-----	-----

Q. What is kernel launch?

Answer:

Calling GPU function from CPU.

DEEP LEARNING SIDE

Q. Why deep learning instead of machine learning?

Answer:

Deep learning automatically learns complex features from large datasets.

Q. Difference between ANN and DNN?

ANN	DNN
-----	-----

Few hidden layers	Multiple hidden layers
-------------------	------------------------

Q. Why hidden layers are required?

Answer:

To learn complex non-linear patterns.

Q. What are weights and biases?

Answer:

Parameters learned during training to adjust predictions.

Q. What is neuron?

Answer:

Basic computational unit in neural network.

Q. What is dropout?

Answer:

Technique that randomly disables neurons during training to reduce overfitting.

Q. Why sigmoid not used in hidden layers?

Answer:

Sigmoid can cause vanishing gradient problem.

Q. Why ReLU popular?

Answer:

Simple, fast and reduces vanishing gradient issue.

Q. Difference between supervised and unsupervised learning?

Supervised	Unsupervised
------------	--------------

Uses labels	No labels
-------------	-----------

Q. Why training and testing datasets separated?

Answer:

To evaluate model generalization on unseen data.

Q. What is hyperparameter?

Answer:

Parameters manually set before training like epochs, learning rate, batch size.

Q. What is learning rate?

Answer:

Controls how much weights change during optimization.

Q. What happens if learning rate is too high?

Answer:

Training becomes unstable and may not converge.

Q. What happens if learning rate is too low?

Answer:

Training becomes very slow.

VERY IMPORTANT DEMONSTRATION QUESTIONS

These are EXTREMELY common.

“Show where parallelism is happening.”

OpenMP

Show:

```
#pragma omp parallel for
```

CUDA

Show:

```
__global__
```

CNN

Show:

Conv2D

RNN

Show:

LSTM / SimpleRNN

MOST IMPORTANT “WHY THIS MODEL?” QUESTIONS

Why CNN for images?

Answer:

CNN extracts spatial image features efficiently.

Why RNN for stock prediction?

Answer:

RNN handles sequential dependencies in time-series data.

Why DNN for Boston housing?

Answer:

DNN can model complex relationships between housing features and prices.

Why IMDB uses binary classification?

Answer:

Only two sentiment classes exist.

POSSIBLE OUTPUT QUESTIONS

“Why validation accuracy is less than training accuracy?”

Answer:

Because model performs better on seen training data than unseen validation data.

“What if validation loss increases?”

Answer:

Possible overfitting.

“What if both accuracies are low?”

Answer:

Possible underfitting.

COMMON ERROR QUESTIONS

Since you faced many, externals sometimes ask these.

Q. Why shape mismatch occurs?

Answer:

Input feature dimensions do not match model input layer dimensions.

Q. Why NameError occurs?**Answer:**

Variable/module used before declaration or import.

Q. Why NotFittedError occurs?**Answer:**

Prediction attempted before training model.

GENERAL THEORY QUESTIONS FOR DEEP LEARNING + HIGH PERFORMANCE COMPUTING**(Questions externals ask apart from practicals)**

These are VERY commonly asked to check whether you actually understand the subjects beyond execution.

HIGH PERFORMANCE COMPUTING (HPC)**Basics****Q1. What is High Performance Computing (HPC)?****Answer:**

HPC is the use of parallel processing and supercomputers to solve complex computational problems efficiently.

Q2. Why HPC is needed?**Answer:**

To solve large and computationally intensive problems faster.

Q3. Applications of HPC?**Answer:**

- Weather forecasting
 - Scientific simulations
 - AI/Deep Learning
 - Bioinformatics
 - Space research
 - Cryptography
-

Q4. What is parallel computing?

Answer:

Executing multiple computations simultaneously using multiple processors.

Q5. Difference between sequential and parallel computing?

Sequential	Parallel
-------------------	-----------------

One task at a time	Multiple tasks simultaneously
--------------------	-------------------------------

Q6. What is concurrency?**Answer:**

Managing multiple tasks at overlapping time periods.

Q7. Difference between concurrency and parallelism?

Concurrency	Parallelism
--------------------	--------------------

Tasks overlap	Tasks truly execute simultaneously
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Architecture Questions**Q8. What is Flynn's Classification?****Types:**

- SISD
 - SIMD
 - MISD
 - MIMD
-

Q9. Explain SIMD.**Answer:**

Single Instruction Multiple Data executes same instruction on multiple data simultaneously.

Q10. Explain MIMD.**Answer:**

Multiple processors execute different instructions on different data.

Q11. What is multicore processor?**Answer:**

Processor containing multiple CPU cores.

Q12. Difference between core and processor?**Processor****Core**Physical CPU chip Execution unit inside CPU

Memory Questions**Q13. Difference between shared and distributed memory?****Shared Memory****Distributed Memory**Common memory Separate memory per processor

Q14. What is cache memory?**Answer:**Small high-speed memory used to reduce access time.

Q15. What is memory hierarchy?**Fastest → Slowest**

Registers

Cache

RAM

Secondary Storage

Performance Questions**Q16. What is speedup?**

$$Speedup = \frac{Sequential\ Time}{Parallel\ Time}$$

Q17. What is efficiency?

$$Efficiency = \frac{Speedup}{Processors}$$

Q18. What is scalability?**Answer:**

Ability to improve performance with increasing processors.

Q19. What is throughput?

Answer:

Amount of work completed per unit time.

Q20. What is latency?

Answer:

Time delay before execution/output begins.

Q21. Explain Amdahl's Law.

$$Speedup = \frac{1}{(1 - P) + \frac{P}{N}}$$

Where:

- P = parallel portion
 - N = processors
-

Q22. What does Amdahl's Law state?

Answer:

Maximum speedup is limited by sequential portion of program.

OpenMP Questions

Q23. What is OpenMP?

Answer:

API for shared-memory parallel programming.

Q24. Advantages of OpenMP?

Answer:

- Simple syntax
 - Easy parallelization
 - Portable
 - Shared memory support
-

Q25. What is pragma directive?

Answer:

Compiler instruction for parallel execution.

Q26. What is thread?

Answer:

Lightweight execution unit.

Q27. What is synchronization?

Answer:

Coordination between threads.

Q28. What is race condition?

Answer:

Multiple threads modifying shared data simultaneously causing incorrect results.

Q29. What is deadlock?

Answer:

Processes waiting indefinitely for resources.

Q30. What is load balancing?

Answer:

Equal workload distribution among processors.

CUDA Questions

Q31. What is CUDA?

Answer:

NVIDIA's GPU parallel programming platform.

Q32. Difference between CPU and GPU?

CPU

GPU

Few powerful cores Thousands of smaller cores

Q33. What is kernel in CUDA?

Answer:

GPU-executed function.

Q34. What is thread block?

Answer:

Group of CUDA threads.

Q35. What is GPU memory hierarchy?

Answer:

Registers → Shared Memory → Global Memory

DEEP LEARNING

Basics

Q36. What is Artificial Intelligence?

Answer:

Simulation of human intelligence in machines.

Q37. What is Machine Learning?

Answer:

Systems learn patterns from data without explicit programming.

Q38. What is Deep Learning?

Answer:

Subset of ML using multi-layer neural networks.

Q39. Difference between AI, ML and DL?

AI

ML

DL

Broad field Subset of AI Subset of ML

Neural Network Basics

Q40. What is neural network?

Answer:

Computational model inspired by biological neurons.

Q41. What is neuron?

Answer:

Basic computational unit performing weighted calculations.

Q42. What are weights?

Answer:

Parameters controlling importance of inputs.

Q43. What is bias?

Answer:

Additional parameter shifting activation output.

Q44. What is activation function?

Answer:

Function introducing non-linearity into neural network.

Q45. Why non-linearity important?

Answer:

Real-world data relationships are non-linear.

Activation Functions

Q46. What is ReLU?

$$ReLU(x) = \max(0, x)$$

Q47. Advantages of ReLU?

Answer:

- Faster training
 - Reduces vanishing gradient
-

Q48. What is sigmoid?

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Q49. Why sigmoid for binary classification?

Answer:

Outputs probabilities between 0 and 1.

Q50. What is softmax?

Answer:

Activation used for multiclass classification.

Training Questions

Q51. What is forward propagation?

Answer:

Passing inputs through network to generate predictions.

Q52. What is backpropagation?

Answer:

Updating weights using prediction errors.

Q53. What is gradient descent?

Answer:

Optimization algorithm minimizing loss function.

Q54. What is optimizer?

Answer:

Algorithm updating model weights.

Q55. What is Adam optimizer?

Answer:

Adaptive optimization algorithm combining momentum and RMSProp.

Q56. What is epoch?

Answer:

One complete pass through training data.

Q57. What is batch size?

Answer:

Number of samples processed together before weight update.

Q58. What is learning rate?

Answer:

Controls size of weight updates.

Loss and Accuracy

Q59. What is loss function?

Answer:

Measures prediction error.

Q60. What is Mean Squared Error?

$$MSE = \frac{1}{n} \sum (y - \hat{y})^2$$

Q61. What is Binary Cross Entropy?

$$L = -[y \log(p) + (1 - y) \log(1 - p)]$$

Q62. What is accuracy?

Answer:

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions}$$

Model Problems

Q63. What is overfitting?

Answer:

Model memorizes training data but performs poorly on unseen data.

Q64. What is underfitting?

Answer:

Model fails to learn patterns properly.

Q65. How to reduce overfitting?

Answer:

- Dropout
 - More data
 - Regularization
 - Early stopping
-

CNN Questions

Q66. What is CNN?

Answer:

Neural network specialized for image processing.

Q67. What is convolution?

Answer:

Applying filters to extract features.

Q68. What is pooling?

Answer:

Reducing feature map dimensions.

RNN Questions

Q69. What is RNN?

Answer:

Neural network for sequential data.

Q70. What is LSTM?

Answer:

Improved RNN handling long-term dependencies.

Q71. What is vanishing gradient problem?

Answer:

Gradients become extremely small during backpropagation.

NLP Questions

Q72. What is NLP?

Answer:

Natural Language Processing.

Q73. What is tokenization?

Answer:

Splitting text into tokens/words.

Q74. What is embedding?

Answer:

Dense vector representation of words.

MOST IMPORTANT FINAL QUESTIONS

Q75. Why GPU important in Deep Learning?**Answer:**

Deep learning requires massive parallel computations efficiently handled by GPUs.

Q76. Relation between HPC and Deep Learning?**Answer:**

Deep learning training requires HPC techniques like parallel computing and GPU acceleration.

Q77. Why parallelism important in AI?**Answer:**

Training deep neural networks involves huge computations requiring faster execution.

Q78. Real-life applications of Deep Learning?**Answer:**

- Face recognition
- Chatbots
- Self-driving cars
- Medical diagnosis
- Recommendation systems